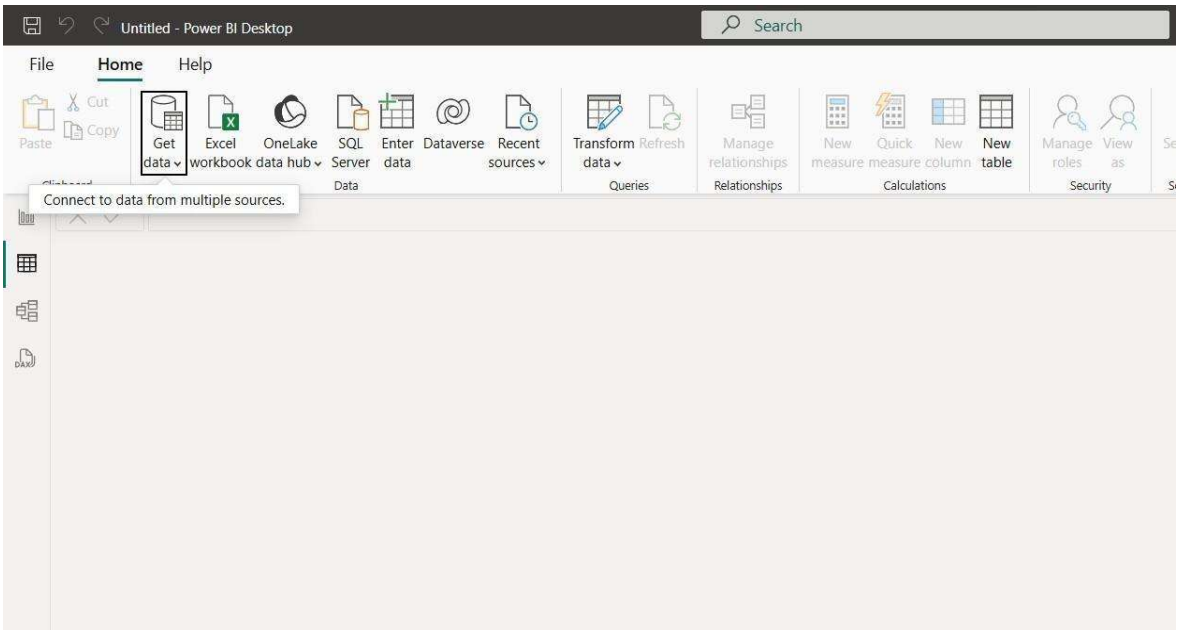
EX.NO: 10 (a)	DATA ANALYSIS USING TIME SERIES ANALYSIS
DATE:	

AIM:

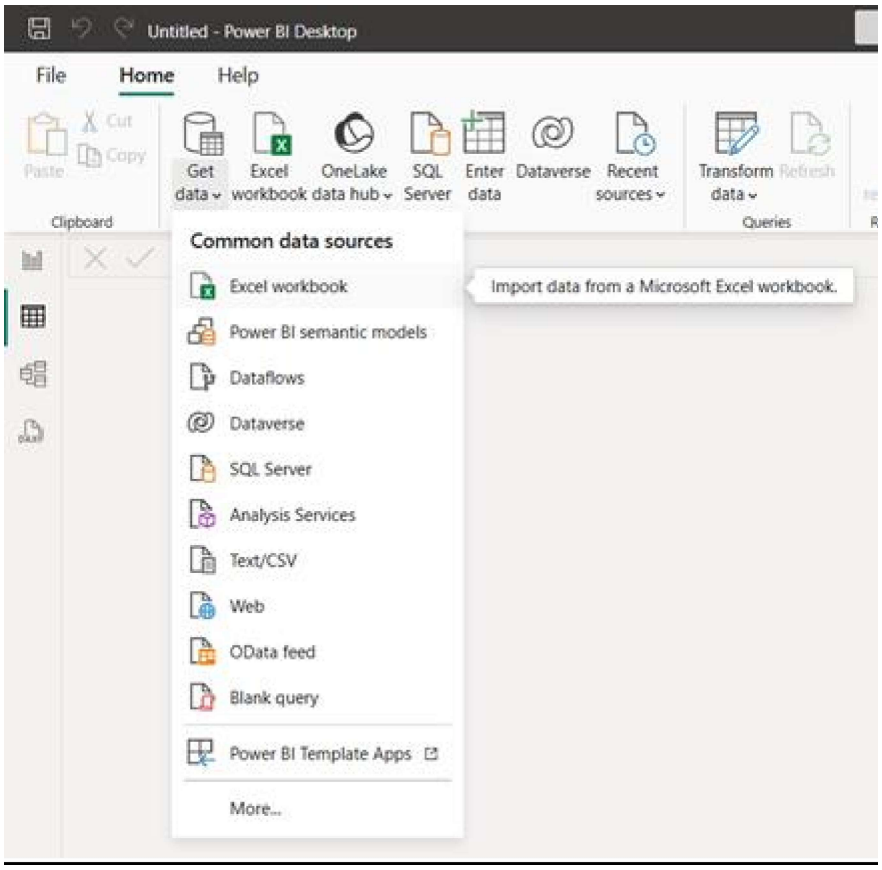
To Create a Dashboard Analytics on Real-Time Sales Using Time Series Dataset

PROCEDURE:

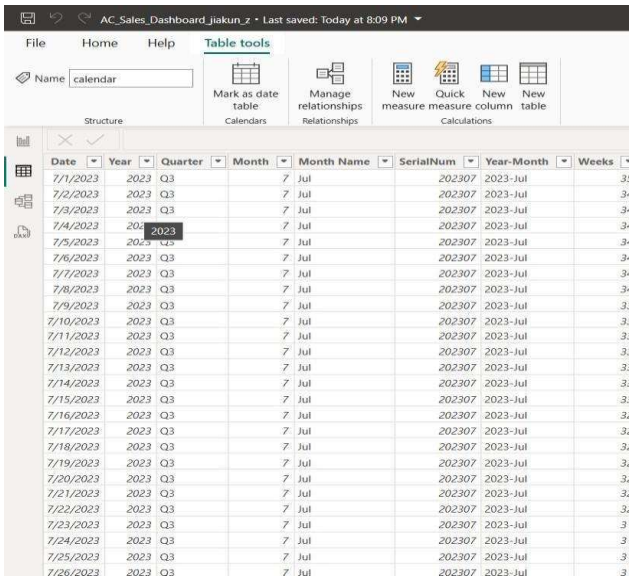
Step1: To begin creating a dashboard for real-time sales analytics using the chocolate sales dataset in Power BI, start by clicking on "Get data" in the ribbon at the top of the screen. This will allow you to import your dataset into Power BI for further analysis and visualization.



Step 2: After clicking "Get data" in Power BI Desktop, navigate to the dropdown menu that appears. From the list of common data sources, select "Excel workbook" to proceed with importing your chocolate sales dataset. This option is highlighted in the menu, allowing you to connect to your Excel file containing the dataset. Power BI will then guide you to locate and load your file for further analysis.



Step 3:

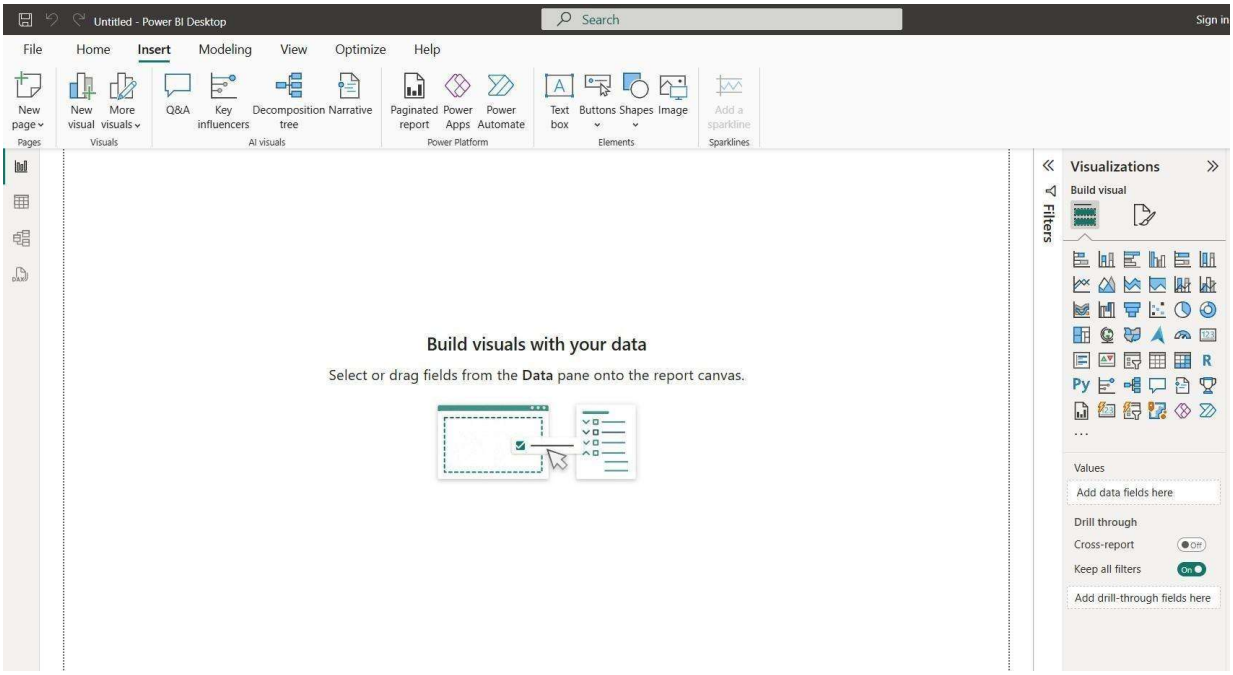


The image shows the Power BI Desktop interface with a table view of a dataset, likely part of your chocolate sales dataset, displayed in the "Fields" pane on the right. The table includes columns such as "Date," "Year," "Quarter," "Month," "Month Name," "SerialNum," and "Year-Month," with data entries for various dates in 2023. This indicates that you've already imported your dataset (e.g., the Excel workbook with chocolate sales data) and are now viewing or preparing it for analysis. The ribbon at the top includes options like "Home," "Insert," "View," and "Modeling," which you can use to transform data, create relationships, or build visualizations. The "Table tools" tab is active, suggesting you're working with a specific table, and you can use features like "Mark as date table," manage relationships, or create new measures and columns.

Next Step:

To proceed with creating a dashboard for real-time sales analytics using the time series dataset, click on "Insert" in the ribbon at the top of the screen, then select "New visual" to start adding visualizations (e.g., line charts, bar charts) to analyze sales trends over time, such as chocolate sales by date or month.

Step 4:

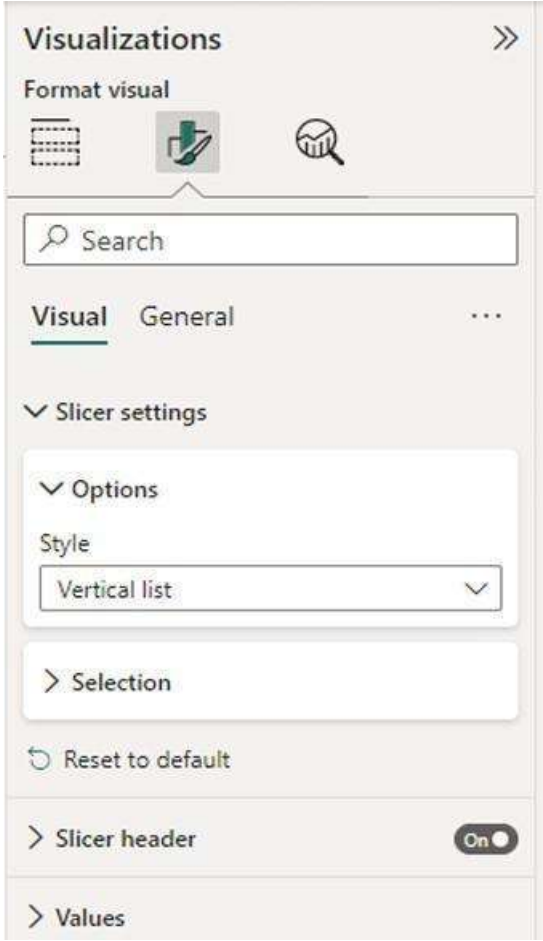


The next step was to click on "Insert" in the ribbon, then select "New visual" to start adding visualizations for your chocolate sales dataset. The current image shows Power BI Desktop with a blank report canvas and the "Visualizations" pane open on the right, indicating you've already followed that step and are now ready to create a specific visualization. The canvas prompts you to "Build visuals with your data" by selecting or dragging fields from the "Fields" pane (not visible in this image but typically on the right) onto the report canvas.

From the "Visualizations" pane on the right, select the line chart icon (a chart with lines and dots) to create a time series visualization for real-time sales analytics, such as tracking chocolate sales over time using the "Date" or "Year-Month" fields from your dataset. Then, drag the appropriate fields (e.g., "Date" to the Axis and "Sales" or a similar metric to the Values) from the "Fields" pane onto the line chart to visualize sales trends.

Step 5:

Next step was to select the line chart icon from the "Visualizations" pane and drag fields like "Date" and "Sales" from your chocolate sales dataset to create a time series visualization for real-time sales analytics. The current image shows the "Format visual" pane in Power BI, specifically for a slicer visual, with options like "Slicer settings," "Style" (set to "Vertical list"), "Selection," "Slicer header," and "Values." This indicates you've already added a slicer to your report, likely to filter data (e.g., by date, product, or region), and are now customizing its appearance or behavior.



To enhance your dashboard for real-time sales analytics, go back to the "Visualizations" pane, select the line chart icon again (if not already present), and ensure it's placed on the report canvas. Then, drag the "Date" field to the Axis and a sales-related field (e.g., "Sales Amount" or a similar metric) to the Values to create or refine the time series visualization of chocolate sales trends, allowing you to analyze data over time while using the slicer to filter interactively.

Step 6:

The image shows the "Fields" section of a combo chart (line and column chart) in the Power BI "Visualizations" pane, with "Segment" assigned to the X-axis, "Sum of Sales" to the Column y-axis, and "Country" to Small multiples. This configuration suggests you're creating a visualization to analyze chocolate sales data, comparing sales across segments (e.g., product types or categories) and breaking them down by country. To integrate this into your report for real-time sales analytics using the time series dataset, follow these steps:

1. Verify the Data Fields: Ensure the fields "Segment," "Sum of Sales," and "Country" are correctly pulled from your chocolate sales dataset in the "Fields" pane. If needed, adjust by dragging the appropriate fields (e.g., replace "Segment" with "Date" or "Year-Month" for a time series focus, and ensure "Sales" reflects your sales metric).

2. Adjust for Time Series: To align with real-time sales analytics, modify the X-axis to use a time-based field like "Date" or "Year-Month" instead of "Segment." Drag the "Date" or "Year-Month" field from the "Fields" pane to the X-axis, replacing "Segment," to create a time series visualization of sales trends.

3. Enhance the Visualization: Keep "Sum of Sales" on the Column y-axis to show sales amounts over time. If you want to compare sales across countries, retain "Country" in Small multiples to create separate charts for each country, or move it to the Column legend for a different breakdown.

4. Format the Visual: Click on "Format visual" (next to the chart icon in the "Visualizations" pane) to customize the chart's appearance, such as colors, labels, and titles, to make it clear and visually appealing for your dashboard.
5. Add to the Report Canvas: Drag the combo chart from the "Visualizations" pane onto the report canvas if it's not already there. Position it alongside other visuals (e.g., slicers or line charts) to create a comprehensive dashboard for analyzing chocolate sales trends over time.
6. Interact with Slicers: If you have slicers (e.g., from previous steps), ensure they're connected to filter this chart by dragging relevant fields like "Date" or "Country" into the slicer visual. This allows real-time filtering of the sales data.
7. Save and Publish: Once satisfied with the visualization, save your Power BI file by clicking "File" > "Save" in the top-left corner. To share the dashboard, publish it to Power BI Service by clicking "File" > "Publish" > "Publish to Power BI."

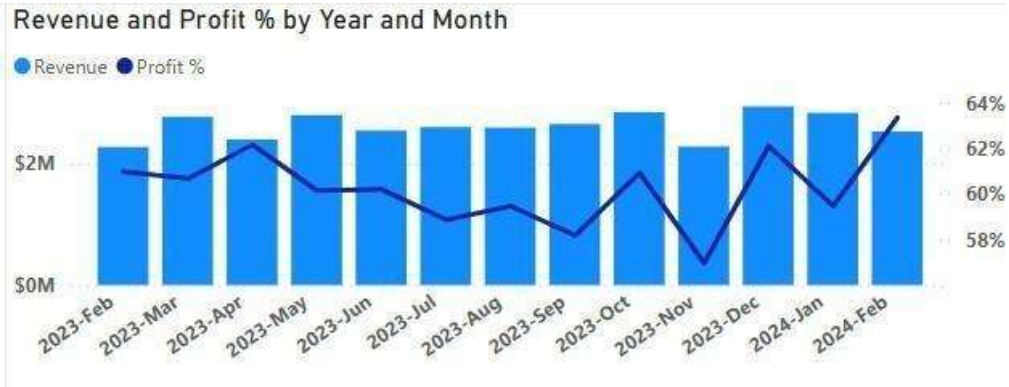
Step 7:

$34M	2M	60.29%	$21M	6113	624
Revenue	Total Boxes	Profit %	Total Profit	Total Transactions	Low-Box Shipments

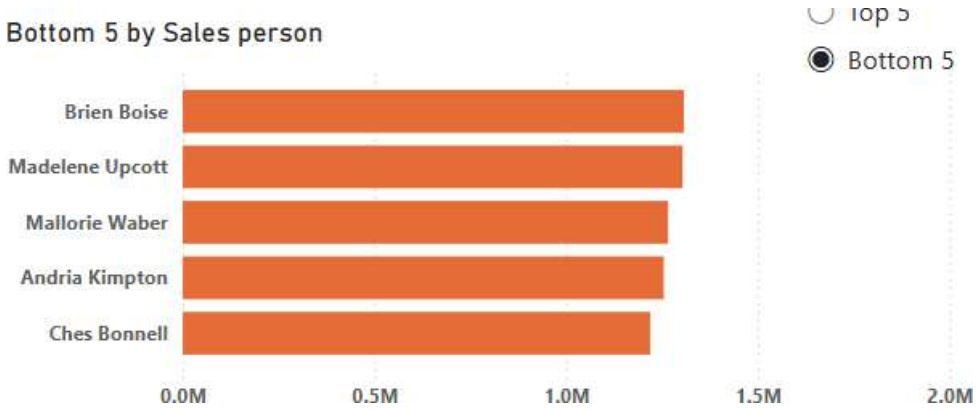
This visualized data in Power BI is likely represented using a Card visualization for each metric (Revenue, Total Boxes, Profit %, Total Profit, Total Transactions, Low-Box Shipments). Each card displays a single key value ($34M, 2M, 60.29%, $21M, 6113, 624) with its corresponding label, arranged horizontally across the report canvas. You can create this in Power BI by selecting the Card icon from the "Visualizations" pane, dragging each metric (e.g., Sum of Revenue, Count of Boxes) from the "Fields" pane into the card, and formatting the cards for alignment and style under "Format visual." Slicers can be added to filter the data dynamically for real-time analysis.

Country	Revenue	Cost	Profit %
Australia	$5,703,536	$2,144,923	62.39%
Canada	$5,725,895	$2,371,919	58.58%
India	$5,648,465	$2,180,480	61.40%
New Zealand	$5,875,218	$2,258,013	61.57%
UK	$5,471,935	$2,339,443	57.25%
USA	$5,617,463	$2,223,664	60.42%
Total	**$34,042,511**	**$13,518,441**	**60.29%**

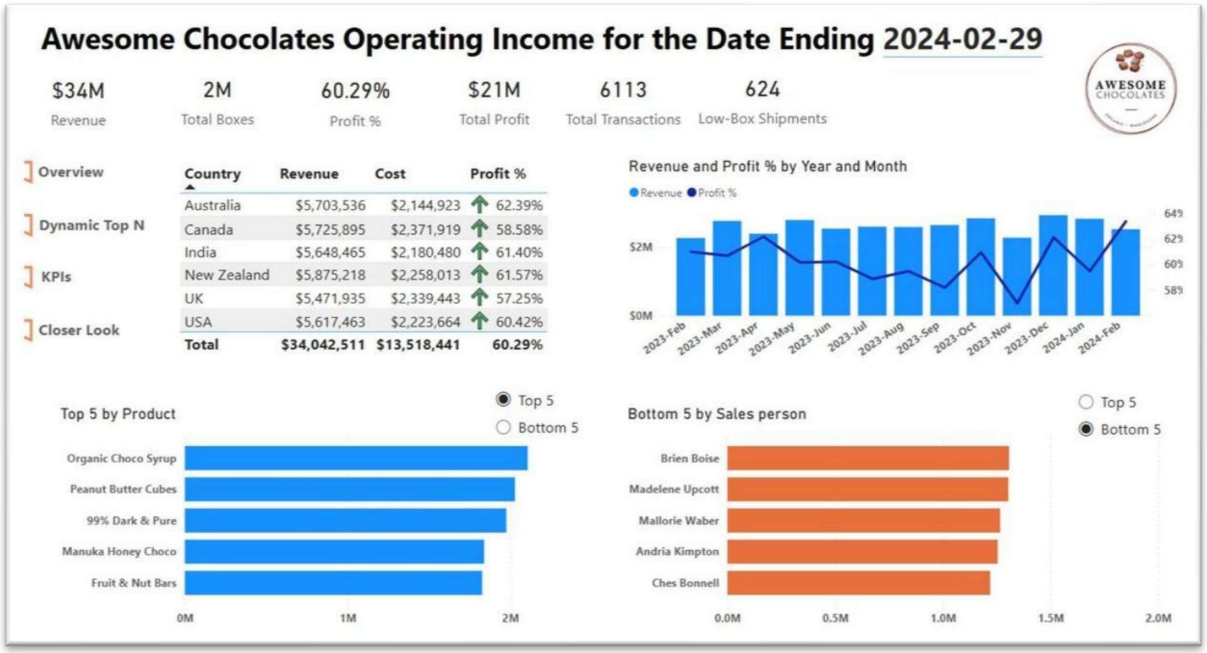
This data is visualized in Power BI using a Table visualization. The table displays columns for "Country," "Revenue," "Cost," and "Profit %," with rows for each country (Australia, Canada, India, New Zealand, UK, USA) and a total row. You can create this by selecting the Table icon from the "Visualizations" pane, dragging the "Country," "Revenue," "Cost," and "Profit %" fields from the "Fields" pane into the table, and formatting it under "Format visual" to add conditional formatting (e.g., green arrows for profit) and totals. Slicers can be added for interactive filtering by country or other fields.



This data is visualized in Power BI using a Combo Chart combining a Column Chart for Revenue and a Line Chart for Profit %. The X-axis represents months from January 2023 to February 2024, the Column y-axis shows Revenue in $M, and the Line y-axis (on the right) displays Profit % (58%-64%). To create this, select the Combo Chart icon in the "Visualizations" pane, drag "Month" to the Axis, "Revenue" to the Column values, and "Profit %" to the Line values, then format under "Format visual" to adjust colors and scales.



The chart displays the "Bottom 5 by Sales person" (Brien Boise, Madelene Upcott, Mallorie Waber, Andria Kimpton, Ches Bonnell) on the y-axis, with sales amounts (0M to 2M) on the x-axis. To create this, select the Bar Chart icon in the "Visualizations" pane, drag "Sales Person" to the Axis and a sales metric (e.g., Sum of Sales) to the Values, sort by sales in ascending order to show the bottom 5, and format under "Format visual" to adjust colors and labels.



This dashboard for Awesome Chocolates Operating Income is visualized in Power BI using a combination of visualizations:

- Card visualizations display key metrics (Revenue: $34M, Total Boxes: 2M, Profit %: 60.29%, Total Profit: $21M, Total Transactions: 6113, Low-Box Shipments: 624).
- A Table visualization under "Country" shows Revenue, Cost, and Profit % by country, with totals.
- A Combo Chart (Column for Revenue, Line for Profit %) tracks trends by year and month.
- Bar Charts highlight the Top 5 Products by sales and Bottom 5 Sales Persons, created by dragging respective fields to the Axis and Values, sorting, and formatting under "Format visual." Slicers or filters can enhance interactivity.

RESULT:

The analysis in Power BI, applied to the Excel chocolate sales data, generates a dynamic dashboard with adjustable Target Profit on Card visuals ($21M + adjustment), a Line Chart showing Profit % trends (58%-64%) with a projected line, a Table updating Country-wise metrics (e.g., Australia: $5.7M, 62.39% + impact), and Bar Charts for Top 5 Products and Bottom 5 Sales Persons. This setup enables real-time simulation of sales target impacts for enhanced decision-making.

Ex.No: 10 (b)	DATA ANALYSIS USING TIME SERIES ANALYSIS
Date:	

AIM:

To perform the time series analyze in POWER BI by using python script and visualize the charts.

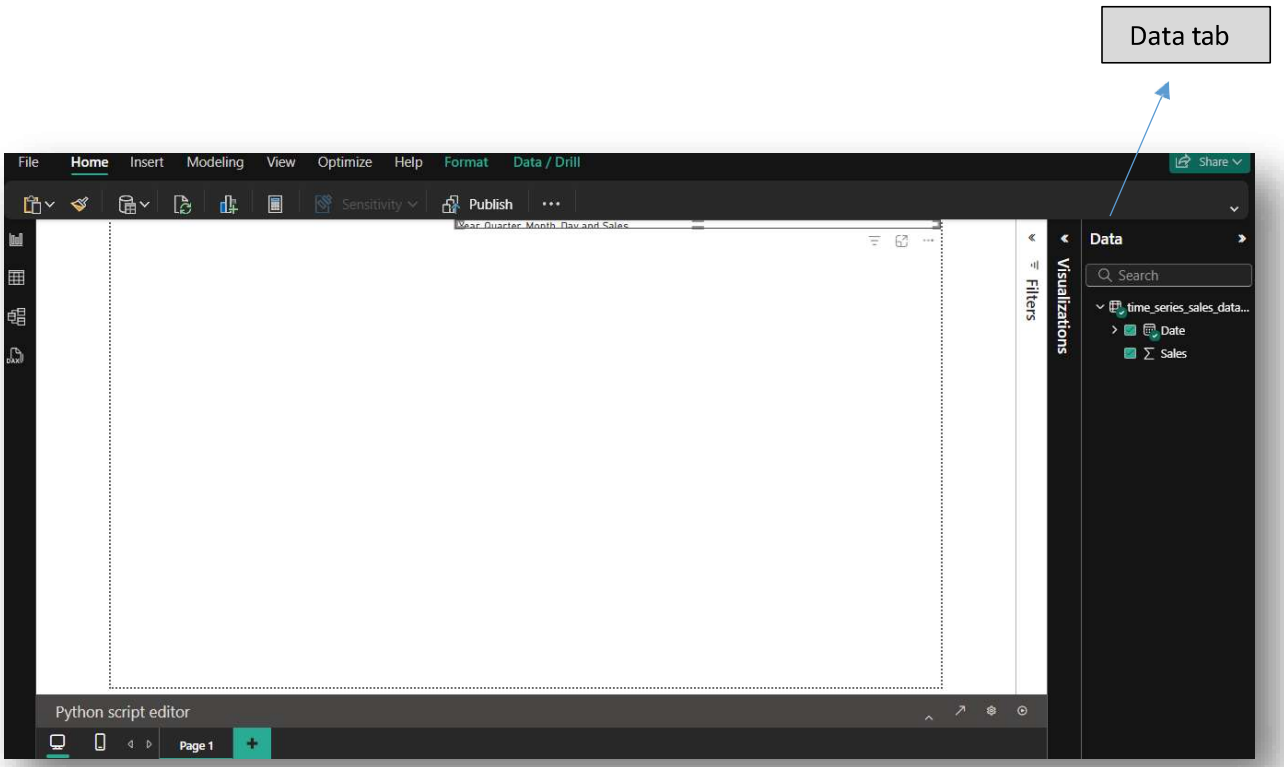
IMPLEMENTATION:

1. Install and Open Power BI

1. **Download & Install Power BI**
   - Visit the official Power BI Download Page.
   - Download and install **Power BI Desktop**.
2. **Open Power BI Application**
   - Launch the Power BI Desktop application after installation.

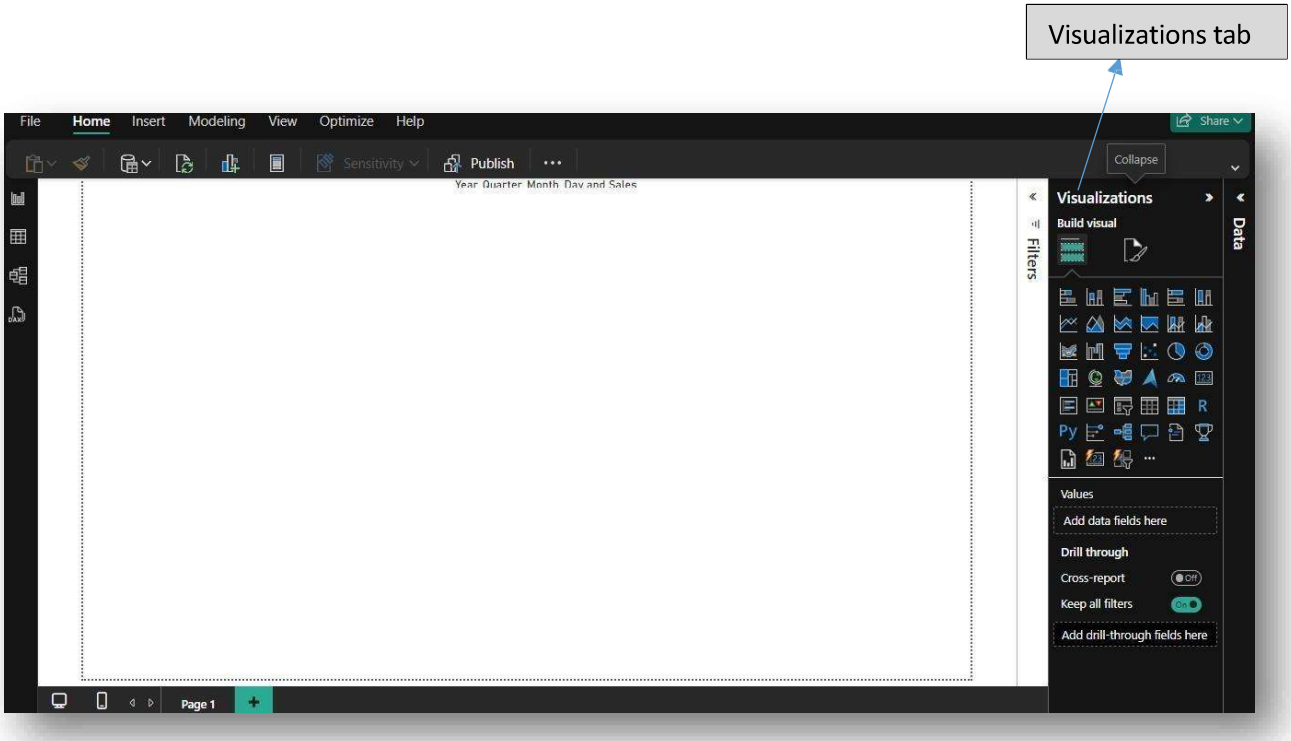
2. Load the Time Series Dataset

3. **Download the Dataset**
   - Ensure you have a **time series dataset** in a suitable format (CSV, TXT, etc.).
   - Example dataset: **"time_series_sales_dataset.csv"**.
4. **Import the Dataset into Power BI**
   - Open **Power BI Desktop**.
   - Click **Home > Get Data > Text/CSV**.
   - Select **"time_series_sales_dataset.csv"** and click **Load**.
   - Ensure that the **Date column** is formatted as **Date/Time**.
5. **Verify the Imported Data**
   - Click the **Data View** option (right panel) to preview the dataset.



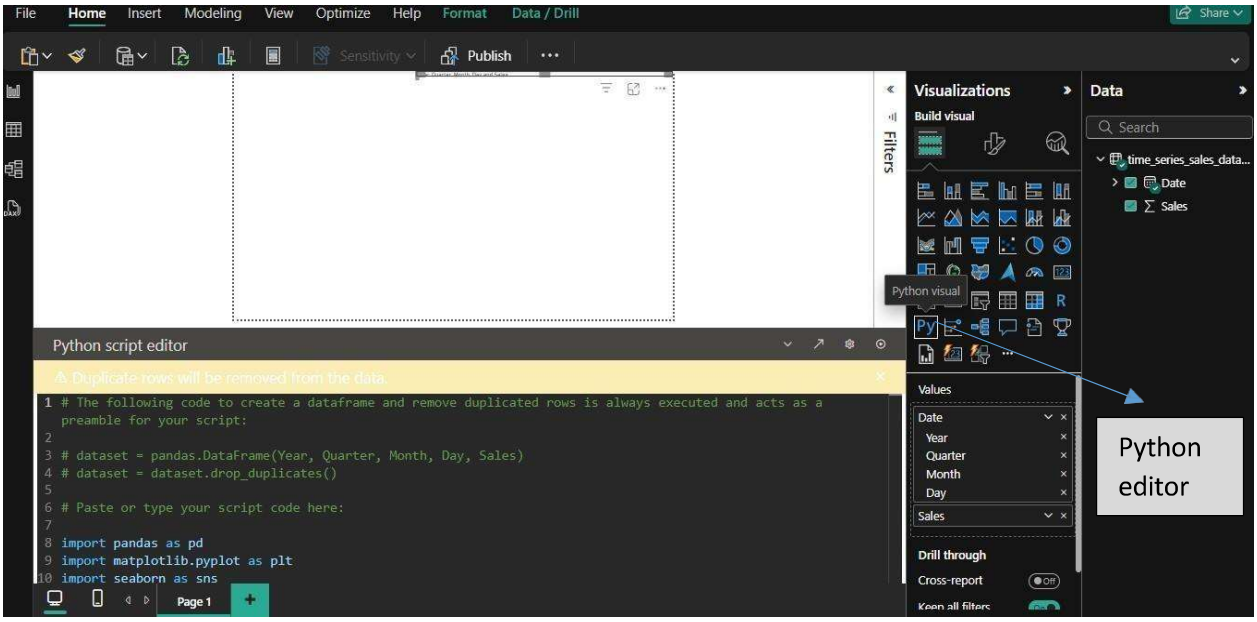
3. Visualize Time Series Data in Power BI

6. **Open the Visualization Pane**
   - Click the **Report View** tab.
   - Click the **"Visualization"** pane to see all available visualizations.
7. **Create a Time Series Line Chart (optional)**
   - From **Visualizations**, select the **Line Chart**.
   - **Drag Date** to the **X-axis**.
   - **Drag Sales** to the **Y-axis**.
   - Title the chart as **"Time Series Sales Analysis"**.



4. Run Python Script for Advanced Analysis

8. **Enable Python Scripting in Power BI**
   - Click **File > Options and Settings > Options**.
   - Under **Python scripting**, set the Python home directory (Anaconda/Miniconda).
9. **Add Python Script in Power BI**
   - Click **"Python Visual" (Py icon)** in the **Visualization Pane**.
   - Select the required **columns** from the dataset.

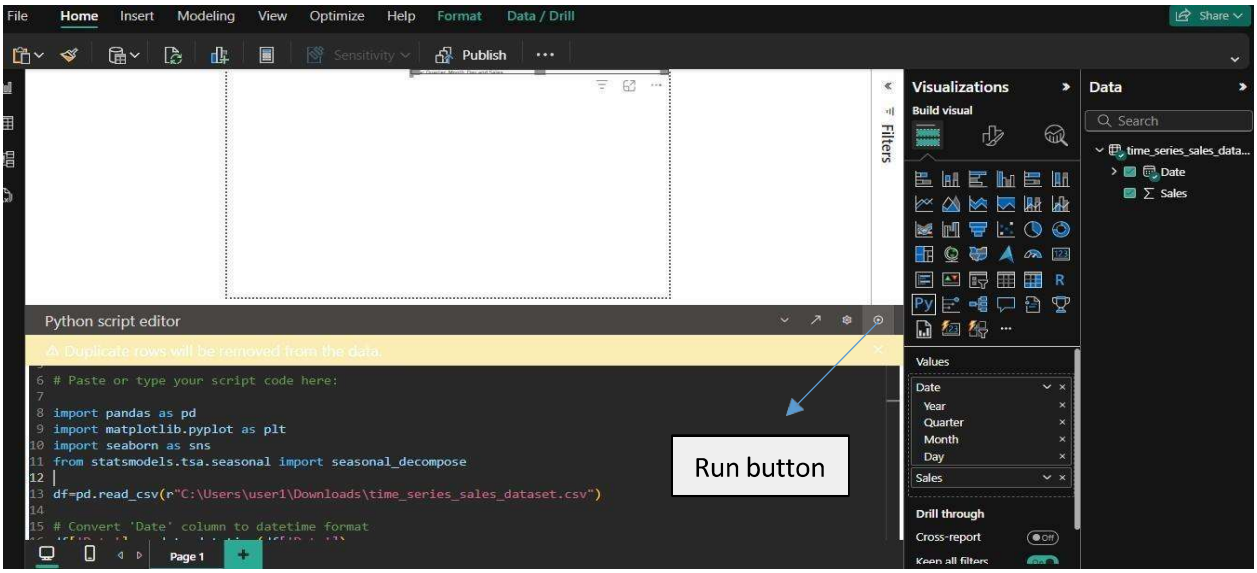


10. **Paste and Run the Python Script**

- Open the **Python Script Editor** in Power BI.
- Paste the following **Python script**:

11. **Run the Python Script**

- Click the **Run button** on the top of the Python script editor.
- Wait for execution to complete.



CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.seasonal import seasonal_decompose

df=pd.read_csv(r"C:\Users\user1\Downloads\time_series_sales_dataset.csv")

# Convert 'Date' column to datetime format
df['Date'] = pd.to_datetime(df['Date'])

# Set 'Date' as the index (Required for time series analysis)
df.set_index('Date', inplace=True)

# Perform Seasonal Decomposition (Additive Model)
decomposition = seasonal_decompose(df['Sales'], model='additive', period=30)  #
Assuming monthly seasonality

# Extract trend, seasonal, and residual components
df['Trend'] = decomposition.trend
df['Seasonality'] = decomposition.seasonal
df['Residual'] = decomposition.resid

# Create visualizations
plt.figure(figsize=(12, 8))

# Original Sales Data
plt.subplot(4, 1, 1)
plt.plot(df.index, df['Sales'], label='Original Sales', color='blue')
plt.title('Original Sales Data')
plt.legend()

# Trend Component
plt.subplot(4, 1, 2)
plt.plot(df.index, df['Trend'], label='Trend', color='red')
plt.title('Trend Component')
plt.legend()
```

```
# Seasonal Component
plt.subplot(4, 1, 3)
plt.plot(df.index, df['Seasonality'], label='Seasonality', color='green')
plt.title('Seasonality Component')
plt.legend()

# Residual Component
plt.subplot(4, 1, 4)
plt.plot(df.index, df['Residual'], label='Residual', color='purple')
plt.title('Residual Component')
plt.legend()

plt.tight_layout()
plt.show()
```

OUTPUT:

RESULT:

The program was executed successfully.